PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Chapter OL-T41 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

3 classified items • 16 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T41 • Expertise Q16+ • 3 items • 16 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Bearings, Scale Drawing & Constructions

Unit 2 • Expertise • 16 marks • 3 item(s)

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-1H Q18 demand **5 marks**

18 The diagram shows the positions of three villages, A, B and C

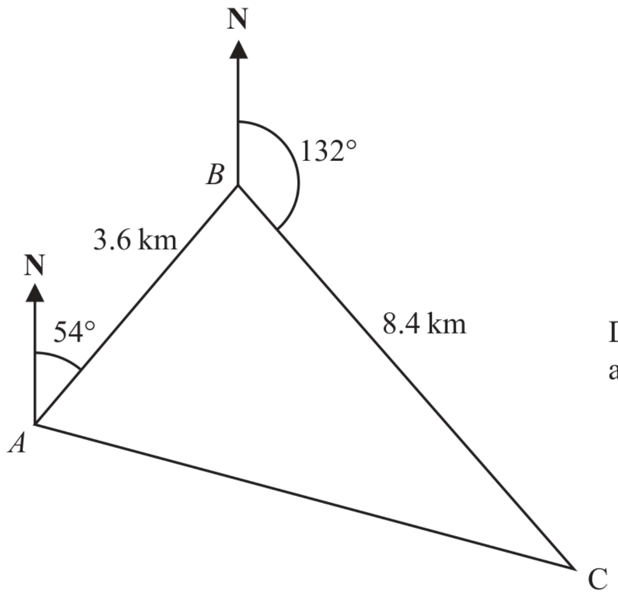


Diagram **NOT** accurately drawn

The bearing of B from A is 054°
The bearing of C from B is 132°

Melur walks from A to B
She then walks from B to C and from C to A

Melur walks at an average speed of 6 km/h

Work out the total time Melur takes.
Give your answer in hours and minutes.

.... hours .... minutes

WORKING SPACE

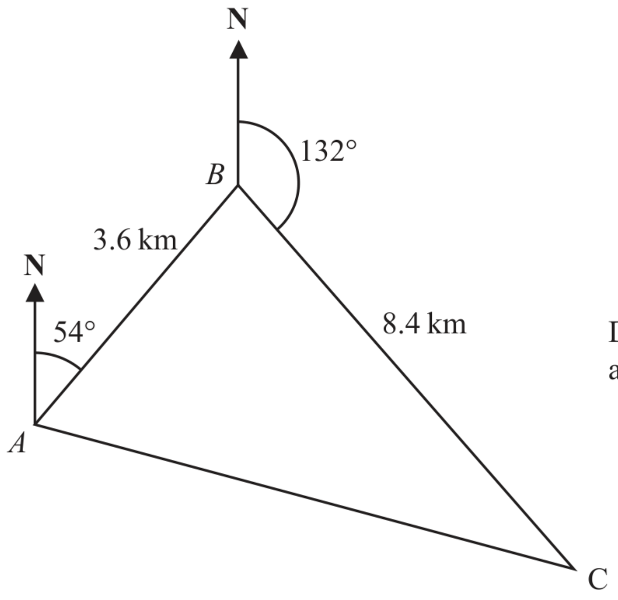
Continue your method

5 marks

WORKING SPACE

4MA1-1H Q18 demand **5 marks**

18 The diagram shows the positions of three villages, *A*, *B* and *C*



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Give your answer in hours and minutes.

.... hours .... minutes

Bearings and total walking time **5 marks**

Demand – Bearings and total walking time

Find angle ABC

$$\angle ABC = 54^\circ + (180^\circ - 132^\circ) = 102^\circ.$$

Find AC

$$AC^2 = 3.6^2 + 8.4^2 - 2(3.6)(8.4)\cos 102^\circ \Rightarrow AC \approx 9.802\text{ km}.$$

Find time

$$t = \frac{3.6 + 8.4 + 9.802}{6} = 3.6337\text{ h} = 3\text{ h } 38\text{ min (approximately)}.$$

Final answer

3 hours 38 minutes

4MA1-2H Q24 M01 **6 marks**

24 A, B and C are three points on horizontal ground.

$AB = 8.4$ metres $BC = 9.2$ metres

B is on a bearing of $067°$ from A
C is on a bearing of $129°$ from B

Calculate the bearing of A from C
Give your answer correct to the nearest degree.

4MA1-2H Q24 M01 **6 marks**

....

WORKING SPACE

Find a bearing using the cosine and sine rules **6 marks**

M01 – Find a bearing using the cosine and sine rules

Find the included angle

$$\angle ABC = 67^\circ + (180^\circ - 129^\circ) = 118^\circ.$$

Find AC

$$AC^2 = 8.4^2 + 9.2^2 - 2(8.4)(9.2)\cos 118^\circ \Rightarrow AC = 15.09\ldots.$$

Find an angle

Use the sine rule to obtain the angle at C, about 29–30°.

Convert to a bearing

The bearing of A from C is 280° (nearest degree).

Final answer

280 degrees

16 The diagram shows the positions of three ships, A, B and C.

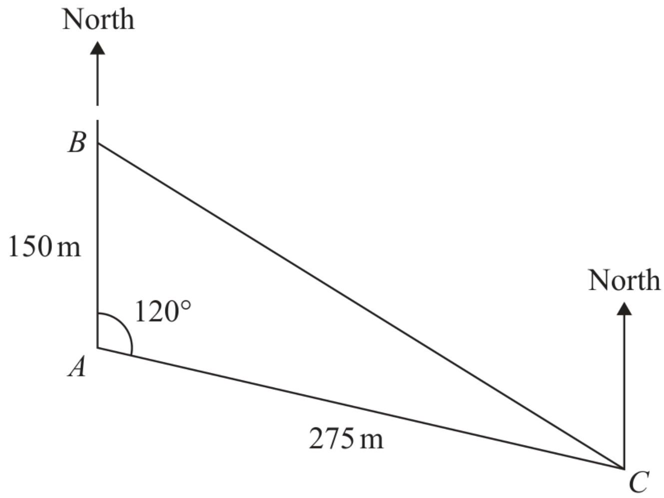

Diagram **NOT** accurately drawn

Ship B is due north of ship A.

The bearing of ship C from ship A is 120°

Calculate the bearing of ship C from ship B.
Give your answer correct to the nearest degree.

....°

Worked solution

January 2021 | Question 16 | 5 marks

Family: Bearing geometry with trigonometry • Variant: Sine or cosine rule supports a final bearing

Method / working

1. Use the cosine rule: BC^2=150^2+275^2-2(150)(275)cos120°.
2. BC=373.4… m.
3. Use the sine rule to find the angle at A or C.
4. The required bearing from B is 120°+20.3°~140.3°.
5. To the nearest degree, the bearing is 140°.

Final answer: 140°